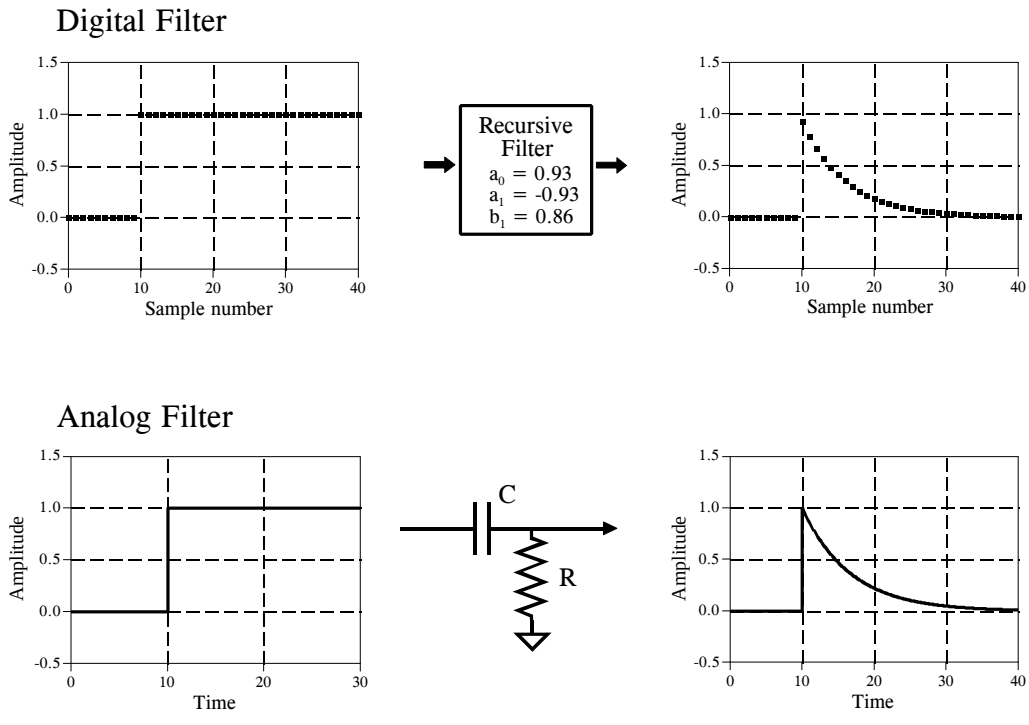

FIGURE 19-3
Single pole high-pass filter. Proper coefficient selection can also make the recursive filter mimic an electronic RC high-pass filter. These single pole recursive filters can be used in DSP just as you would use RC circuits in analog electronics.

to execute, and produce few surprises. The coefficients are found from these simple equations:

EQUATION 19-2
Single pole low-pass filter. The filter's response is controlled by the parameter, *x*, a value between zero and one.

$$a_0 = 1 - x$$
$$b_1 = x$$

EQUATION 19-3
Single pole high-pass filter.

$$a_0 = (1 + x)/2$$
$$a_1 = -(1 + x)/2$$
$$b_1 = x$$

The characteristics of these filters are controlled by the parameter, *x*, a value between zero and one. Physically, *x* is the amount of *decay* between adjacent samples. For instance, *x* is 0.86 in Fig. 19-3, meaning that the value of each sample in the output signal is 0.86 the value of the sample before it. The higher the value of *x*, the slower the decay. Notice that the